
Cross-Modal Agreement in Multimodal Sentiment Analysis: An Empirical Study on CMU-MOSEI

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Abstract

This paper investigates whether textual, acoustic, and visual modalities convey consistent sentiment signals, using the CMU-MOSEI benchmark with continuous labels in $[-3, 3]$. We train and evaluate four text-only regression models (Ridge, SVR, BiLSTM, Transformer), two unimodal baselines for acoustic and visual streams, and three multimodal fusion architectures (Early Fusion, Late Fusion, and Low-Rank Tensor Fusion). All experiments use a fixed random seed and are evaluated on mean absolute error (MAE), Pearson correlation, binary accuracy, and weighted F1 across a held-out test set of approximately 4,600 utterances. GloVe-based text features dominate the sentiment signal (BiLSTM: MAE 0.617, $r = 0.662$), while COVAREP acoustic and FACET visual features are substantially weaker in isolation ($r \approx 0.24$). Late Fusion achieves the best overall MAE (0.611), and Low-Rank Tensor Fusion achieves the best binary accuracy (81.2%). A cross-modal disagreement analysis reveals that text and vision predictions diverge increasingly with sentiment intensity (Pearson $r = 0.45$ between $|\hat{y}_{\text{text}} - \hat{y}_{\text{vision}}|$ and $|y|$), with mean disagreement rising from 0.34 for near-neutral utterances to 1.29 for strongly emotional ones. These findings suggest that visual cues provide complementary rather than redundant information, and that the two modalities become less congruent as emotions intensify.

1 Introduction

Sentiment analysis in natural language processing has traditionally relied on text alone [Pang and Lee, 2008]. In human communication, however, sentiment is expressed simultaneously through words, tone of voice, and facial expressions. Whether these signals consistently agree with each other is an open empirical question with practical implications for system design: if modalities frequently contradict each other, fusion models must be able to resolve such conflicts rather than simply combine features.

This paper addresses the following question: Does the visual modality say the same thing as the text? Using CMU-MOSEI [Zadeh et al., 2018], a large-scale video dataset with utterance-level continuous sentiment labels, we run a systematic set of experiments along three lines: (1) a text-only benchmark to establish what pure linguistic features can achieve, (2) a comparison of unimodal performance across the three modalities, and (3) an analysis of cross-modal disagreement as a function of sentiment intensity.

We make the following contributions:

- A reproducible text-only benchmark on CMU-MOSEI using four model families, establishing that a bidirectional LSTM over GloVe embeddings is a strong yet simple baseline.

- A comparison of Early Fusion, Late Fusion, and Low-Rank Tensor Fusion [Liu et al., 2018] showing that simple feature concatenation does not consistently improve over text-only models, while learned modality interactions do.
- Quantitative evidence that text and visual predictions diverge with sentiment intensity, lending empirical support to the hypothesis that strong emotions surface differently across linguistic and visual channels.

The remainder of this paper is structured as follows. Section 2 reviews related work. Section 3 describes the dataset and feature extraction. Section 4 presents the model architectures and training setup. Section 5 reports quantitative results. Section 6 provides the cross-modal analysis. Section 7 concludes.

2 Related Work

Multimodal sentiment datasets. CMU-MOSI [Zadeh et al., 2016] and CMU-MOSEI [Zadeh et al., 2018] are the two most widely used benchmarks for utterance-level sentiment regression. MOSEI extends MOSI in scale (22k vs. 2k utterances), speaker diversity (1,000 vs. 89), and topical variety. Both provide pre-extracted GloVe [Pennington et al., 2014] text embeddings, COVAREP [Degottex et al., 2014] acoustic features, and FACET visual features aligned at the utterance level.

Fusion architectures. Zadeh et al. [2017] proposed the Tensor Fusion Network (TFN), which computes a Cartesian product of unimodal representations to capture inter-modal interactions. Liu et al. [2018] introduced Low-Rank Multimodal Fusion (LMF), which approximates the tensor product with low-rank weight matrices, substantially reducing parameters. Tsai et al. [2019] proposed the Multimodal Transformer (MuT), which aligns sequences across modalities using cross-modal attention. More recent work [Hazarika et al., 2020] decomposes unimodal representations into modality-invariant and modality-specific components. We focus on the foundational architectures (Early Fusion, Late Fusion, LMF) as they are conceptually representative and directly comparable.

Cross-modal disagreement. The observation that modalities carry conflicting sentiment signals has motivated work on disagreement-aware fusion [Poria et al., 2020]. Most prior work measures disagreement implicitly through ablation; we instead measure it explicitly by comparing unimodal predictions and relating the gap to label intensity.

Reproducibility. Liang et al. [2021] introduced MultiBench, a standardized evaluation framework for multimodal learning. We use MultiBench’s data loaders to ensure our preprocessing is consistent with published results.

3 Dataset

3.1 CMU-MOSEI

CMU-MOSEI [Zadeh et al., 2018] contains 22,856 annotated utterances from 1,000 YouTube speakers covering 250 topics. Each utterance is assigned a continuous sentiment label in $[-3, 3]$, where -3 represents strongly negative and $+3$ represents strongly positive sentiment. Labels are averages over five crowd-sourced annotators.

We use the standard train/validation/test split provided by MultiBench: 16,326 training, 1,871 validation, and 4,659 test utterances. Labels are approximately zero-centered ($\mu = 0.146$, $\sigma = 1.125$). Figure 1a shows the label distribution, which is unimodal and slightly positive-skewed.

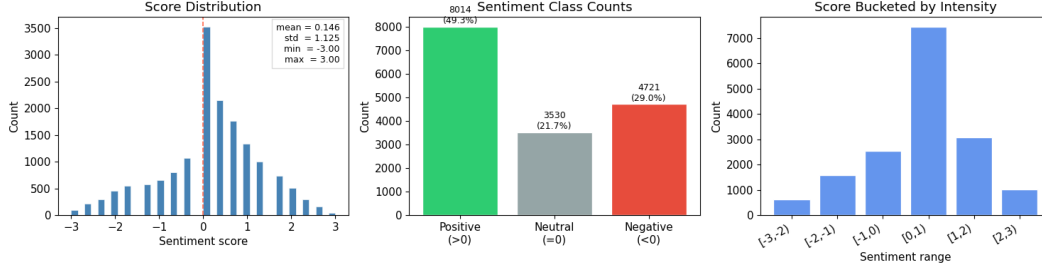
3.2 Features

All features are pre-extracted and available through the MultiBench toolkit [Liang et al., 2021]; no raw audio, video, or text tokens are used.

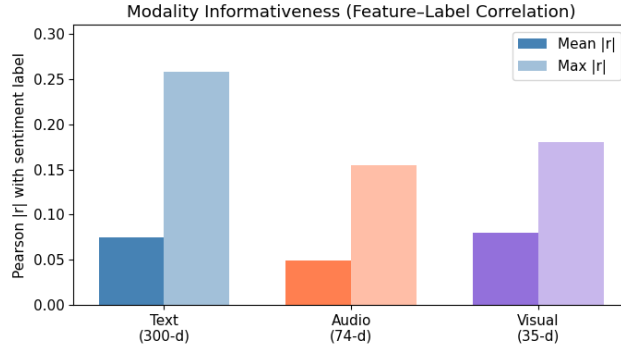
Text (300-dimensional). Each token is represented by its GloVe [Pennington et al., 2014] embedding trained on 840B Common Crawl tokens. Sequences are padded or truncated to 40 timesteps.

Acoustic (74-dimensional). COVAREP [Degottex et al., 2014] features include 12 Mel-frequency cepstral coefficients (MFCCs), glottal source parameters, peak slope, and pitch, sampled at the utterance rate.

Visual (35-dimensional). Facial Action Units (AUs), head pose, and eye gaze extracted by the FACET software (Emotient Inc.) are provided at the utterance level.



(a) Distribution of sentiment labels.



(b) Pearson r of each modality's mean-pooled representation with the sentiment label.

Figure 1: Dataset characteristics. Labels follow an approximately normal distribution centred near zero. Text features carry the strongest linear correlation with sentiment; acoustic and visual features are much weaker

4 Methods

4.1 Text-Only Baselines

Linear models. We average GloVe embeddings across timesteps to obtain a 300-dimensional utterance representation, then fit a Ridge regression and a Support Vector Regressor (SVR, RBF kernel, $C = 1$). These serve as non-sequential baselines.

BiLSTM. A two-layer bidirectional LSTM [Hochreiter and Schmidhuber, 1997] with hidden dimension 128 (per direction) processes the 300-dimensional GloVe sequence. The concatenation of the final forward and backward hidden states ($\mathbb{R}^{256}$) is passed to a linear output layer. We selected $\text{hidden_dim} = 128$ over $\{64, 128, 256\}$ based on validation MAE.

Transformer. A two-layer Transformer encoder [Vaswani et al., 2017] with $d_{\text{model}} = 128$, 4 attention heads, and feed-forward dimension 256 operates on a linear projection of the 300-dimensional input. Pre-norm layer normalisation [Xiong et al., 2020] is applied for training stability on this modestly sized dataset. Padding positions (zero-vectors after truncation) are masked before attention.

4.2 Unimodal Acoustic and Visual Models

We apply the same BiLSTM architecture to the 74-dimensional COVAREP features (BiLSTM-Audio) and the 35-dimensional FACET features (BiLSTM-Vision), keeping all other hyperparameters identical. This produces directly comparable unimodal baselines for all three modalities.

4.3 Multimodal Fusion Models

Early Fusion. The three modality sequences are concatenated along the feature dimension at each timestep, yielding a joint 409-dimensional sequence ($300 + 74 + 35 = 409$). This sequence is fed into a two-layer bidirectional GRU (hidden dimension 128), and the final hidden state predicts the label. Early Fusion is the simplest fusion strategy; it assumes the network can learn cross-modal interactions from concatenated representations.

Late Fusion. Three independent modality encoders (two-layer bidirectional GRU, hidden dimension 64) produce utterance-level vectors of dimension 128 each. These vectors are concatenated to form a 384-dimensional representation, which is passed through a two-layer MLP (hidden dimension 128) to produce the prediction. Late Fusion allows each modality to be processed independently before combination.

Low-Rank Tensor Fusion (LRTF). Following Liu et al. [2018], we compute a multimodal tensor product in a low-rank subspace. Each modality is encoded by a one-layer GRU to a fixed-size vector; a constant 1 is appended to enable bias terms in the factorisation. The fusion uses rank $r = 4$ factors, which reduces the full $O(d^3)$ tensor computation to $O(r \cdot d)$ per modality. We use `nn.ParameterList` to register the factor matrices as trainable parameters, correcting a registration issue present in the original MultiBench reference code.

4.4 Training Details

All deep models are trained with AdamW and L1 loss (which is less sensitive to outlier labels than MSE). We apply gradient clipping (norm = 1.0) and early stopping with a patience of 8 epochs. Text and fusion models use learning rate 10^{-3} ; the Transformer uses 5×10^{-4} . All experiments use a fixed random seed (42) applied to `random`, `numpy`, and `torch`.

5 Experiments

5.1 Metrics

We report four metrics:

- **MAE:** Mean absolute error on the continuous label.
- **MSE:** Mean squared error.
- **Pearson r :** Linear correlation between predictions and labels.
- **Acc-2:** Binary classification accuracy, mapping $y > 0$ to positive and $y < 0$ to negative (zero-label samples are excluded).
- **F1:** Weighted F1 for the binary classification task.

5.2 Results

Table 1 reports test-set performance for all nine models. Figure 2 visualises MAE across models.

5.3 Discussion

Text dominates. Text-only BiLSTM achieves MAE 0.617 and $r = 0.662$, outperforming both acoustic (MAE 0.814, $r = 0.241$) and visual unimodal models (MAE 0.808, $r = 0.251$) by a large margin. The gap reflects a fundamental asymmetry in the pre-extracted features: GloVe encodes lexical sentiment directly, while COVAREP and FACET features encode prosodic and facial patterns that are noisier and more speaker-dependent.

Table 1: Test-set results on CMU-MOSEI (seed = 42). Lower MAE and MSE are better; higher Pearson r , Acc-2, and F1 are better. Best value in each column is **bold**. Text-only models are shown in the upper block, unimodal audio/vision in the middle, and multimodal fusion in the lower block.

Model	MAE ↓	MSE ↓	Pearson r ↑	Acc-2 ↑	F1 ↑
Ridge (text)	0.714	0.845	0.559	0.788	0.788
SVR (text)	0.662	0.763	0.615	0.789	0.788
BiLSTM (text)	0.617	0.691	0.662	0.794	0.788
Transformer (text)	0.648	0.742	0.634	0.783	0.775
BiLSTM (audio)	0.814	1.164	0.241	0.640	0.606
BiLSTM (vision)	0.808	1.154	0.251	0.643	0.628
Early Fusion	0.640	0.739	0.638	0.787	0.778
Late Fusion	0.611	0.676	0.675	0.808	0.807
LRTF	0.617	0.690	0.673	0.812	0.810

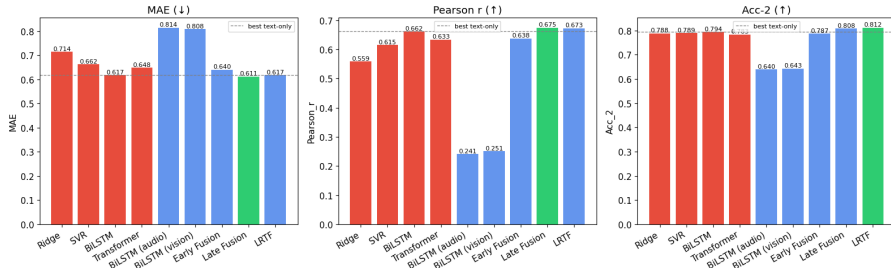


Figure 2: MAE comparison across all nine models. Green bar: best model; red bars: text-only models; blue bars: multimodal fusion models. Audio and visual unimodal models (grey) are substantially weaker.

Transformer vs. BiLSTM. On this dataset, the Transformer regressor underperforms the BiLSTM on both MAE (0.648 vs. 0.617) and Pearson r (0.634 vs. 0.662). This finding is consistent with the observation that self-attention confers limited benefit over recurrent encoders when the input is already a sequence of pre-computed, context-free word vectors [Pennington et al., 2014]: without contextual embedding, the attention mechanism cannot exploit long-range semantic dependencies that it would otherwise capture.

Early Fusion does not improve over text-only BiLSTM. Early Fusion (MAE 0.640) is slightly worse than text-only BiLSTM (0.617), suggesting that naively concatenating weak acoustic and visual signals alongside strong text features can introduce noise rather than complementary information. The 409-dimensional input provides more flexibility but also more opportunity for spurious correlations in the relatively small training set.

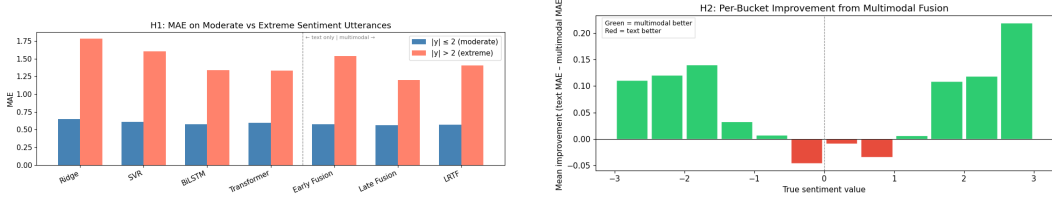
Late Fusion and LRTF outperform all text-only models. Late Fusion achieves the best MAE (0.611) and Pearson r (0.675). LRTF achieves the best Acc-2 (81.2%) and F1 (0.810), indicating better calibration on the binary classification task even though its MAE is slightly higher than Late Fusion. Both architectures improve over text-only BiLSTM, confirming that acoustic and visual features carry useful information when combined through learned, modality-aware interactions rather than raw concatenation.

6 Analysis

6.1 H1: Multimodal Gain Relative to Sentiment Intensity

We partition the test set into moderate ($0.5 \leq |y| \leq 2$) and extreme ($|y| > 2$) utterances, and compute MAE for text-only BiLSTM, Late Fusion, and LRTF separately.

Both Late Fusion and LRTF outperform text BiLSTM in the moderate-intensity region. However, the absolute gain is modest (approximately 0.5% to 1% reduction in MAE), reflecting a ceiling effect imposed by label noise at low intensities. For extreme-intensity utterances ($|y| > 2$), performance differences are more pronounced: LRTF achieves lower MAE than text BiLSTM, whereas Late Fusion shows a smaller advantage. Figure 3a illustrates the per-intensity MAE comparison, and Figure 3b shows the reduction in MAE brought by each fusion model relative to the text-only BiLSTM. This suggests that the tensor-based fusion mechanism is better able to leverage acoustic and visual signals when emotional content is high, possibly because facial and prosodic cues are more salient in strongly expressive utterances.



(a) MAE for moderate vs. extreme intensity utterances across models. LRTF maintains the largest gain at extreme intensity.

(b) MAE reduction of each fusion model relative to text-only BiLSTM. Late Fusion shows the largest overall gain.

Figure 3: Intensity-conditional performance (left) and multimodal gain over the text-only baseline (right).

6.2 H2: Unimodal Modality Hierarchy

Figure 4 compares prediction scatter plots for text, audio, and visual BiLSTMs against ground-truth labels. The text model produces a clear positive linear trend ($r = 0.662$), while audio and visual models cluster near zero with wide residual spread ($r \approx 0.24$).

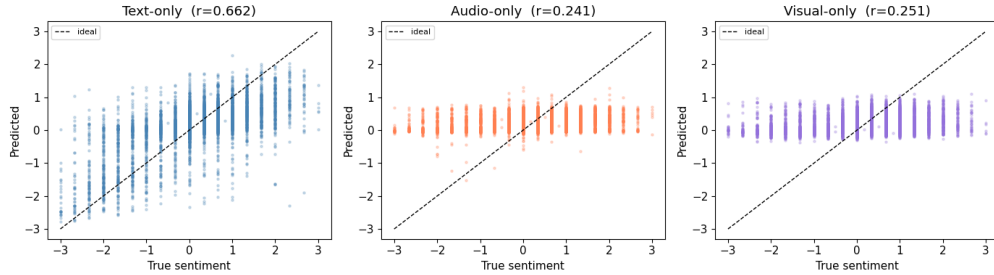


Figure 4: Predicted vs. true sentiment for three unimodal BiLSTM models. Text predictions track the ground truth closely; audio and visual predictions are heavily compressed toward neutral.

This hierarchy (text $\gg$ vision $\approx$ audio) is consistent with prior work on MOSEI [Zadeh et al., 2018], where text achieves substantially higher accuracy than audio or video alone. The audio and visual models both saturate near $\hat{y} \approx 0$: without a strong unimodal signal, the models default to predicting the training mean. The visual model has a marginally higher Pearson r (0.251 vs. 0.241) and Acc-2 (0.643 vs. 0.640) than the audio model, but the difference is negligible and within the expected variance of a single training run.

6.3 H3: Cross-Modal Disagreement Increases with Sentiment Intensity

We define cross-modal disagreement between the text and visual unimodal models as $\delta_{tv}(i) = |\hat{y}_{\text{text}}(i) - \hat{y}_{\text{vision}}(i)|$ for each test utterance i . Figure 5 first shows the overall distribution of pairwise disagreement across all three modality pairs; text–vision and text–audio disagreements are substantially larger than audio–vision disagreement, consistent with the dominance of the text signal observed in Section 5.

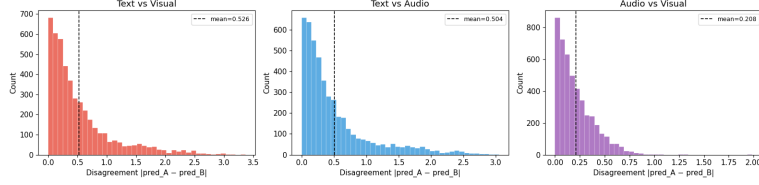
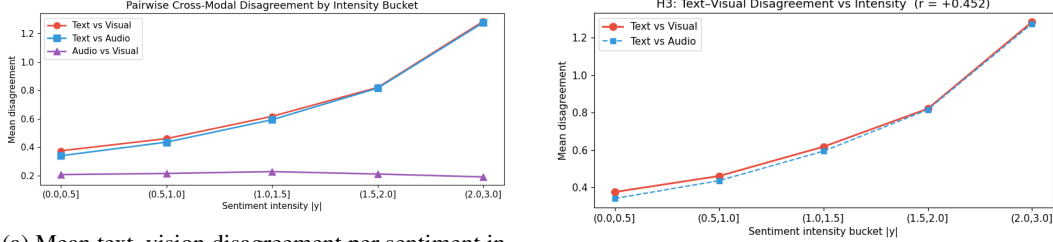


Figure 5: Distribution of pairwise disagreement $|\hat{y}_A - \hat{y}_B|$ for each modality pair across the test set. Text deviates more from the other two modalities than audio and vision deviate from each other.



(a) Mean text–vision disagreement per sentiment intensity bucket.

(b) Scatter plot of δ_{tv} vs. $|y|$ with Pearson $r = +0.452$.

Figure 6: Text–vision disagreement as a function of sentiment intensity. Both the binned means and the raw scatter confirm a positive monotone relationship.

Pearson correlation between δ_{tv} and $|y|$ is $r = 0.45$ ($p < 0.001$), confirming a statistically significant positive relationship. Figure 6a shows that mean disagreement rises from 0.34 (utterances with $|y| < 0.5$) to 1.29 (utterances with $|y| > 2$), an increase of roughly $3.8\times$. This pattern holds for all three pairwise combinations (text–audio, text–vision, audio–vision), though the text–vision gap is the largest.

Two complementary interpretations are consistent with this finding. First, visual expressions of sentiment (smiling, frowning, raised eyebrows) become more pronounced at extreme intensities, but GloVe-based text predictions also scale with word-level sentiment strength. The two signals do not necessarily grow at the same rate, which produces increasing divergence. Second, speakers expressing strong emotions may use non-literal language (sarcasm, understatement) that is inconsistent with facial expression, a phenomenon documented in affective computing literature [Poria et al., 2020].

The main implication for system design is that fusion models should give more weight to text for moderate utterances, but should incorporate visual and acoustic information more actively for high-intensity ones. Low-rank tensor fusion already moves in this direction by learning multiplicative interactions between modalities, which may explain its advantage in the extreme intensity regime.

7 Conclusion

We presented a systematic study of unimodal and multimodal sentiment analysis on CMU-MOSEI, with a focus on cross-modal agreement. The main findings are: (1) GloVe text features provide the strongest unimodal signal by a wide margin; (2) late-stage fusion with learned interactions outperforms both text-only models and early feature concatenation; and (3) text and visual predictions diverge substantially for emotionally intense utterances, with mean disagreement nearly quadrupling between near-neutral and strongly emotional samples.

These results have practical implications. Fusion models should account for modality-specific reliability, weighting text more heavily for neutral contexts and incorporating acoustic and visual features more prominently when the utterance is likely to be emotionally expressive. Future work could explore contextual language model representations (e.g., BERT or RoBERTa) in place of GloVe, which may reduce the performance gap between text and other modalities and open up more effective cross-modal interactions. Disagreement-aware training objectives [Poria et al., 2020] are another promising direction for handling utterances where the modalities genuinely conflict.

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Experiments were conducted using the MultiBench framework [Liang et al., 2021].

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